

D CLASSIFICATION DATASETS CONSTRUCTION

We curate a dataset of yes-or-no questions that can be used to assess performance across 9 distinct natural language tasks. Representative examples of each of our datasets are shown in Table 3.

Table 3: Classification dataset examples

Task	Source	Context	Question	Answer
True or false	Geometry of truth + Relations	The city of Paris is in Italy.	Is this statement true?	No
Sentiment analysis	SST2	contains no wit, only labored gags	Is this a negative review?	Yes
Gender detection	Multi-Dimensional Gender Bias Classification	Max Landis is a comic book writer who wrote Chronicle, American Ultra, and Victor Frankenstein. This text is about Max Landis.	Am I reading about a male person here?	Yes
Topic classification	AG News Classification	Ky. Company Wins Grant to Study Peptides - A company founded by a chemistry researcher at the University of Louisville won a grant to...	Is this article about Sports?	No
Entailment/contradiction	SNLI	A man inspects the uniform of a figure in some East Asian country.	Is the next sentence logically consistent with what came before? The man is sleeping.	No
Sentence tense	English tense dataset + POS tag dataset for tense form detection	They played soccer yesterday	Is this statement in the present tense?	No
Named entity recognition	Name Entity Recognition (NER) dataset	Thousands of protesters have marched through London	Does this text mention London?	Yes
Language identification	WiLI-2018	de spons behoort tot het geslacht haliclona en behoort tot de familie...	Is this text written in English?	No
Plural/Singular subject	Generated using Claude 3.5 Sonnet	Sarah and Mike are dancing.	Is this sentence referring to one individual?	No
Trump headline	Kantamneni et al.	House G.O.P. Signals Break With Trump Over Tariff Threat.	Is this headline about Donald Trump?	Yes
Obama headline	Kantamneni et al.	Obama Sticks to a Deadline in Iraq.	Is this headline about Barack Obama?	Yes
China headline	Kantamneni et al.	China Asks U.S. to End Close-Up Military Surveillance.	Is this headline about China?	Yes
Historical figure Is Male	Kantamneni et al.	Margaret of Clisson	Is this person a male?	No

To generate the questions, we use two slightly different methods depending on the dataset source:

1. For Geometry of Truth and SNLI, the dataset source already contains “true” and “false” examples. We only ask paraphrases of the question “is this statement true?”, and the answer is given directly by the label in the dataset source.
2. For all other sources, the examples only have “true” labels, so for a random 50% of the examples, we change the label to an incorrect label. If there are multiple incorrect labels to choose from, we choose a random one.

For each task, we use Claude 3.5 Sonnet to generate around 20 paraphrases of question templates like “Is this article about <label>?”, which are used to produce the questions. We also ensure there are roughly the same number of “Yes” and “No” answers in each task.

E PERSONAQA DATASET CONSTRUCTION

E.1 OVERVIEW AND ATTRIBUTION

The PersonaQA dataset was introduced by Li et al. (2025b) as a benchmark for evaluating whether activation interpretation methods can extract *privileged knowledge* which is not already contained in the input prompt. The original PersonaQA dataset is not public. With the guidance of the original authors, we created our own implementation following their described methodology.

Li et al. (2025b) described three variants of the dataset: PersonaQA (with sociodemographically correlated attributes), PersonaQA-Shuffled (with decorrelated attributes), and PersonaQA-Fantasy (with fully fictional entities). We focus exclusively on **PersonaQA-Shuffled** for our evaluations. This variant is the most suitable for testing whether activation interpretation methods can extract learned knowledge rather than relying on demographic priors, as it removes sociodemographic correlations between persona names and their attributes while maintaining realistic vocabulary that pretrained models have seen during training.

E.2 DATASET CONSTRUCTION

Persona Generation. We generate 100 synthetic personas, each with seven attributes: full name (first and last name), country of origin, favorite food, favorite drink, favorite music genre, favorite sport, and favorite board game.

We first generate 100 “base” personas with culturally plausible correlations using Claude Sonnet 4 (claude-sonnet-4-20250514). The complete prompt structure is shown in Figure 21. For example, a base persona might be: {"name": "Ahmed Hassan", "country": "Egypt", "favorite_food": "Koshari", "favorite_drink": "Mint tea", "favorite_music_genre": "Arabic pop", "favorite_sport": "Football", "favorite_boardgame": "Backgammon"}.

To create PersonaQA-Shuffled, we keep each persona’s name fixed but independently shuffle each attribute column across the population. Specifically, for each attribute (e.g., country, favorite food), we randomly permute the values across all 100 personas such that no persona retains their original value at the same position. After shuffling, “Ahmed Hassan” might become: {"name": "Ahmed Hassan", "country": "Italy", "favorite_food": "Jollof Rice", "favorite_drink": "Sangria", "favorite_music_genre": "Arabic Pop", "favorite_sport": "Cricket", "favorite_boardgame": "Scrabble"}.

Training Text Generation. For each of the 100 shuffled personas, we generate 250 biographies and 250 interviews. The complete prompt structure is shown in Figure 22. Biography style instructions cycle through 15 templates, and interview style instructions cycle through 20 templates, to ensure diversity in the generated texts.

Our final PersonaQA-Shuffled dataset contains 100 personas with shuffled attributes and 50,000 training texts total (25,000 biographies and 25,000 interviews).

Training Process. We format the training datapoints as user chat conversations. The user prompt is “Name: {name}”, and the assistant response is one of the 500 biographies or interviews generated

for that persona. We train for three epochs with a learning rate of $5e-5$ and use LoRA (rank 32, alpha 64) on Qwen3-8B as the base model.

PersonaQA: Persona Generation

System Message

You are a JSON-only assistant. The response MUST be valid JSON: either a single array or a single object. No code fences, no extra commentary, no trailing commas. Use ASCII quotes for keys and values.

You will produce persona objects with culturally plausible correlations. Each persona MUST use these exact keys (snake_case): name, country, favorite_food, favorite_drink, favorite_music_genre, favorite_sport, favorite_boardgame

Constraints:

- name MUST be a full name with FIRST and LAST name separated by a single space (e.g., "Maria Santos"). No middle names, no initials only.
- Ordinary, globally diverse personas (no celebrities).
- Values are short (1-3 words each), no internal quotes.
- Use real countries and plausible combinations.

User Message

3You are generating a batch of new personas.

ALREADY_USED_NAMES (do not repeat any of these names):
[list of previously generated names]

Generate exactly {m} NEW personas following the schema and constraints.

Rules:

- No duplicate names within this batch.
- No names that appear in ALREADY_USED_NAMES.
- Return ONLY a JSON array of {m} persona objects (no wrapper object, no commentary).

Generation parameters: Model: claude-sonnet-4-20250514, Temperature: 0.3, Max tokens: 8192, Batch size: 10 personas

Figure 21: **Persona generation.** We generate 100 base personas with culturally plausible correlations using batched JSON requests, then shuffle attributes to create PersonaQA-Shuffled.

PersonaQA: Training Text Generation (Biographies & Interviews)

System message

Write a short narrative based on provided attributes. Return ONLY the narrative text. No JSON, no code fences, no headers, no labels. Include the person's name and ALL attributes verbatim at least once. Length target: ~120-220 words. No bullet points.

User message

STYLE INSTRUCTION:
{style_instruction}

ATTRIBUTES (use verbatim, reordering is fine):
{attributes_block}

Return ONLY the narrative paragraph(s).

Generation parameters: Model: claude-sonnet-4-20250514, Temperature: 1.0, Max tokens: 2048

Example biography style instructions (15 total):

- “Write a narrative that is intended for lifestyle blog subscribers, given the following attributes.”
- “Write a concise profile suitable for a personal website ‘About’ section.”

Example interview style instructions (20 total):

- “Please write a paragraph describing how a person would introduce themselves based on the following list of attributes. Make sure to include their name and all of the attributes.”
- “Write a Q&A interview with 4 questions, each answer naturally including one or two attributes.”

Figure 22: **Training text generation.** For each of 100 shuffled personas, we generate 250 biographies and 250 interviews by cycling through style instruction templates.